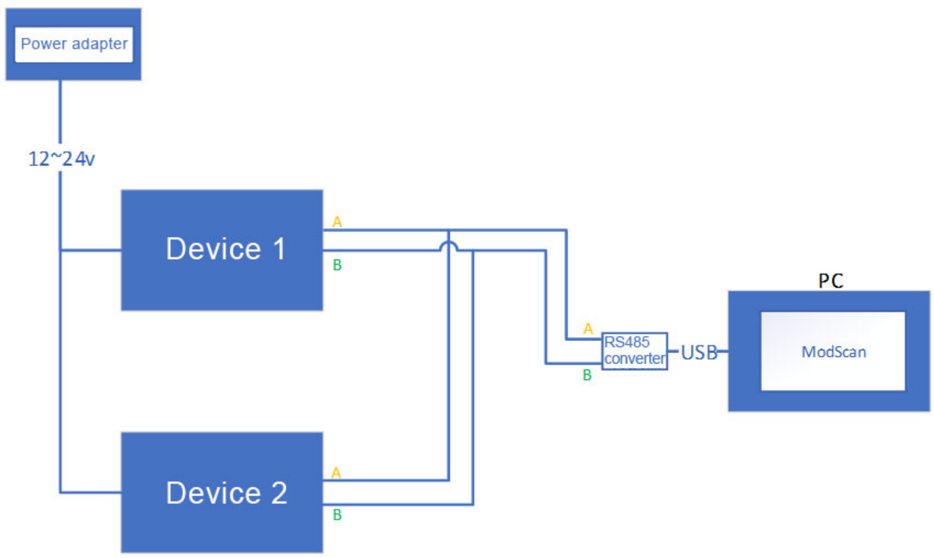
Our company's RS485 communication protocol for all detectors or control panels follows the standard Modbus RTU protocol and can support ModScan master station PC software to read data from specified register addresses. The configuration parameters are as follows:

Precondition:

Wires and connect the detector or control panel first, as shown in the figure:



Picture 1

1.Configure baud rate

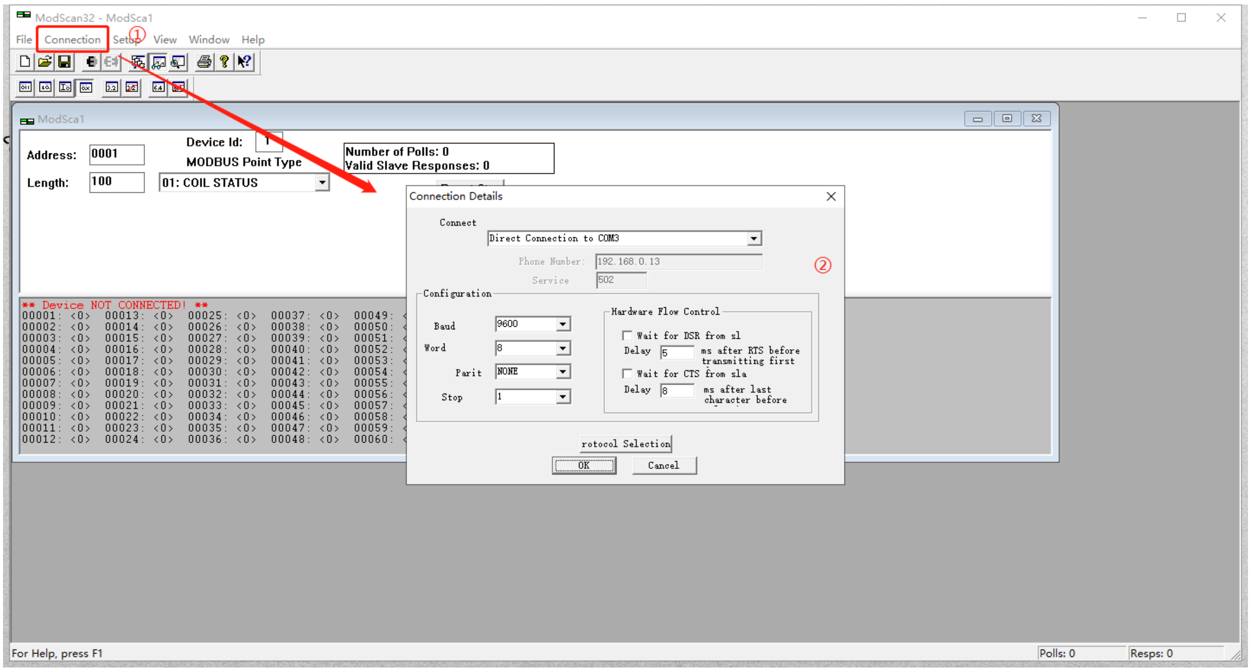
The baud rate of the detector or control panel(slave station) is configured as follows.

If there is a special customization, please contact us.

RS485 communication serial port configuration		
Name	**configuration**	**Remark**
Baud Rate	9600	
Data bit（Word）	8	
Check bit（Parit）	NONE	
Stop bit（Stop）	1	

Table 1

Open the ModScan software, click on "Connection", select the serial port to "Connect", and then use the "Configuration" serial port parameters according to the above icon. Other parameters are default;



Picture 2

2.Configure Modbus parameters

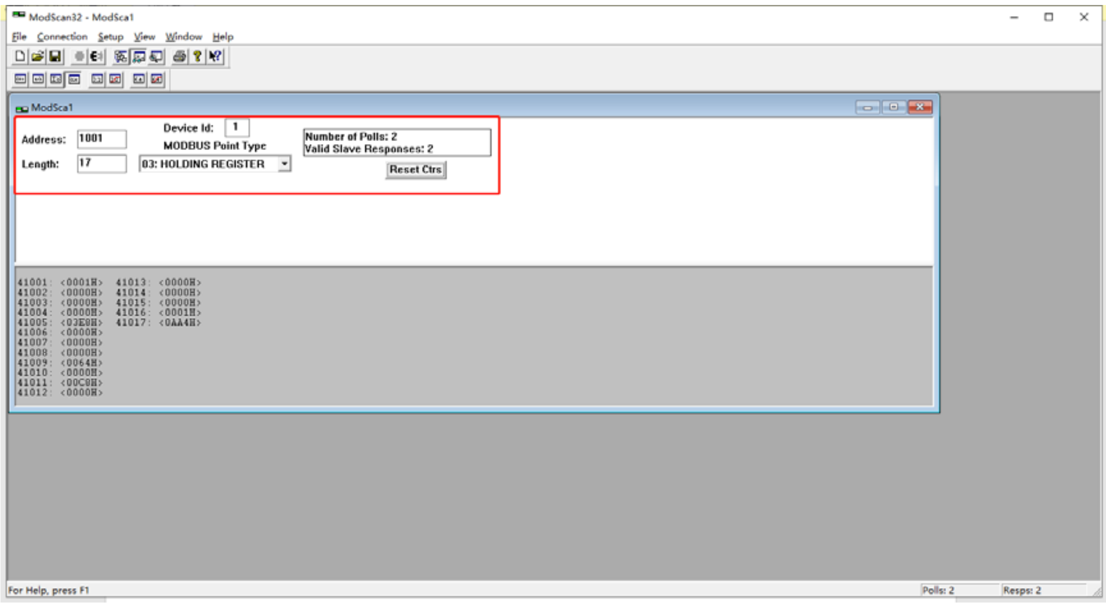
Device Id： According to the device address of the detector/control panel (slave station), if it is a multi (capable of detecting more than 2 gases) detector, the DeviceId of other channels is the device address+channel-1. For example, if the device address/offset address of detector 1 is 1, then the DeviceId of H_2S in channel 2 is 1+2-1=2, and the same applies to other channel algorithms;

Address： It is the register address in the protocol. Here, the 1001 value corresponds to the 41001 register address in the protocol;

Length： The length of the registers that need to be read. It is recommended that the main station start from the 41001 register address and read 17 registers to obtain all gas parameters;

Function code selection "03：HOLDINGREGISTER"

As shown in the following figure



Picture 3

The bottom parameter is the register value returned by the slave station

```
41001: <0001H>  41013: <0000H>
41002: <0000H>  41014: <0000H>
41003: <0000H>  41015: <0000H>
41004: <0000H>  41016: <0001H>
41005: <03E8H>  41017: <0AA4H>
41006: <0000H>
41007: <0000H>
41008: <0000H>
41009: <0064H>
41010: <0000H>
41011: <00C8H>
41012: <0000H>
```

Picture 4

一、Data analysis in the data area

According to Picture 4, the analysis of the register values returned from the slave station is as follows:

Serial NO	**Register address**	**Register value**	**Register Name**	**Numerical content**	**Remark**
1	41001	0x01	Gas type	Check the "Gas Types" table, where 01 represents CO	Communication Protocol Parameter Appendix - Gas Type Definition
2	41002	0x00	Unit	Check the 'Unit' table, 00 represents PPM	Communication Protocol Parameter Appendix - Gas Type Definition
3	41003	0x00	Decimal places	0, indicating Resolution of 0, without decimal point	

4	41004~41005	0x03E8	Measure range	The decimal value of 0x3E8 is 1000, range=register value/10 decimal point Real range:= 1000/100=1000	Assuming the register value is 1000 When the resolution is 0, the true range is 1000/100=1000/1=1000 When the resolution is 1, the true range is 1000/101=1000/10=100.0 When the resolution is 2, the true range is 1000/102=1000/100=10.00 When the resolution is 3, the true range is 1000/103=1000/1000=1000
5	41006~41007	0x00	Gas concentration	Same algorithm as above	
6	41008~41009	0x64	Low alarm value	Same algorithm as above,value is 100	
7	41010~41011	0xC8	High alarm value	Same algorithm as	

				above,value is 200	
8	41012~41013	0x00	STEL alarm value	Same algorithm as above,value is 0	
9	41014~41015	0x00	TWA alarm value	Same algorithm as above,value is 0	
10	41016	0x01	Alarm status	Check the protocol status, 1 indicates' normal '	
11	41017	0x0AA4	ADC	The decimal value of 0x0AA4 is 2724	

Table 2